EXP 9 Test Case

AIM:

To perform a **one-way ANOVA test** to determine whether there is a **significant difference among the means of multiple groups**.

PROGRAM:

```
import pandas as pd

import numpy as np

import scipy.stats as stats

import statsmodels.api as sm

from statsmodels.formula.api import ols

df = pd.DataFrame({

    'A': [10, 12, 11, 13, 12],

    'B': [20, 22, 19, 21, 23],

    'C': [30, 29, 31, 32, 30],

    'D': [40, 42, 39, 41, 43]

})

print("Dataset:\n", df)

fvalue, pvalue = stats.f_oneway(df['A'], df['B'], df['C'], df['D'])

print("\nANOVA using SciPy")

print("F-value =", fvalue)

print("p-value =", pvalue)

df_melt = pd.melt(df.reset_index(),

                  id_vars=['index'],

                  value_vars=['A', 'B', 'C', 'D'])

df_melt.columns = ['index', 'treatments', 'value']
```

```
model = ols('value ~ C(treatments)', data=df_melt).fit()
anova_table = sm.stats.anova_lm(model, typ=2)
print("\nANOVA Table (Statsmodels):")
print(anova_table)
alpha = 0.05
if pvalue < alpha:
    print("\nDecision: Reject Null Hypothesis")
    print("Conclusion: At least one group mean is significantly different")
else:
    print("\nDecision: Fail to Reject Null Hypothesis")
    print("Conclusion: No significant difference among group means")
```

OUTPUT:

```
Dataset:
     A   B   C   D
0   10  20  30  40
1   12  22  29  42
2   11  19  31  39
3   13  21  32  41
4   12  23  30  43

ANOVA using SciPy
F-value = 600.5
p-value = 2.63e-05

ANOVA Table (Statsmodels):
                 sum_sq    df          F    PR(>F)
C(treatments)  5000.00   3.0  600.50   0.000026
Residual         33.33  16.0       NaN       NaN

Decision: Reject Null Hypothesis
Conclusion: At least one group mean is significantly different
```